

PART-B

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	1	7	5	1	14

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

Ans 1. $S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$ $Y_{1 \times 1}$ and $X_{q \times 1}$

To prove:

$r_1^2 = R^2$
 Squared sample first canonical correlation $\rightarrow$ coefficient of determination from multi linear regression of Y on vector X .

r_1^2 is eigenvalue of -

$$S_{yy}^{-1/2} S_{yX} S_{XX}^{-1} S_{Xy} S_{yy}^{-1/2}$$

$S_{yy} = [\text{var}(y)] \Rightarrow S_{yy}^{-1/2} = \frac{1}{\sigma} I$

r_1^2 is eigenvalue of -

$$\frac{1}{|S_{yy}|/n} (S_{yX} S_{XX}^{-1} S_{Xy}),$$

$$[S_{yX}]_{1 \times q} (S_{XX}^{-1})_{q \times q} [S_{Xy}]_{q \times 1} = []_{1 \times 1} \text{ eigenvalue}$$

~~$\sigma^2 = \frac{\sum (y_i - \bar{y})^2}{n}$~~ , as σ^2 is variance of y .

$$S_{yX} = \text{Cov}(y, X)$$

$$= \text{Cov}(X\beta + \epsilon, X)$$

$$= \beta \text{Cov}(X, X) = \beta S_{XX}$$

$$r_1^2 = \frac{\beta S_{XX} S_{XX}^{-1} \cdot \beta S_{XX}/n}{|S_{yy}|/n} = \frac{\beta^2 S_{XX}/n}{|S_{yy}|/n}$$

$$r_1^2 = \frac{\beta^2 S_{xx}}{|S_{yy}|}$$

$$\begin{aligned} \beta &= (x^T x)^{-1} x^T y \\ &= (x^T x)^{-1} x^T (x\beta + \varepsilon) \\ &= \beta + (x^T x)^{-1} x^T \varepsilon \end{aligned}$$

$$SS_{Reg} = \beta^2 S_{xx}$$

$$SS_T = |S_{yy}| \neq$$

$$r_1^2 = \frac{SS_{Reg}}{SS_T} = R^2$$

== hence proved.

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2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix S . The eigenvalues of S are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are e_1, e_2, e_3 , and e_4 . You decide to extract $m = 2$ factors using the principal component method.
- (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{L}$ in terms of the given eigenvalues and eigenvectors.
 - (1 point) Let the first row of $\tilde{L}$ be $\tilde{l}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
 - (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of S .

Ans-2: factor analysis on 4×4 sample covariance matrix S .
 eigenvalues of S : $\lambda_1 = 0.8$ $\lambda_2 = 2.1$ $\lambda_3 = 4.2$ $\lambda_4 = 0.2$
 e_1, e_2, e_3, e_4 are corresponding eigenvectors.
 $m = 2$ factors . using principal component method .

(a) if $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ are eigenvalues of S , and $e_1, e_2, \dots, e_p$ are corresponding eigenvectors,

then: $\tilde{L} = [\sqrt{\lambda_1} e_1 \mid \sqrt{\lambda_2} e_2 \mid \dots \mid \sqrt{\lambda_m} e_m]$

here
 02 $\tilde{L} = [\sqrt{4.2} e_3 \mid \sqrt{2.1} e_2] \approx \text{as } \text{m} = 2$

(b) first row of $\tilde{L} = [0.8, -0.4]$

estimated communality $\tilde{h}_1^2 = \sum_{i=1}^2 l_{1i}^2 = l_{11}^2 + l_{12}^2$

$= (0.8)^2 + (-0.4)^2$

$= 0.64 + 0.16$

$= \underline{\underline{0.8}}$

(c) To prove:

~~$\sum_{i=1}^p \tilde{h}_i^2 = \lambda_1 + \lambda_2$~~

$\sum_{i=1}^p \tilde{h}_i^2 = \underbrace{\lambda_1 + \lambda_2 + \dots + \lambda_m}_{\text{largest } m \text{ eigenvalues}}$

Here: $\sum_{i=1}^4 \tilde{h}_i^2 = \lambda_1 + \lambda_2 = \lambda_3 + \lambda_2$ (for this question)
 $= \underline{\underline{6.3}}$

Proof We know: $e_1 = [e_{11} \ e_{12} \ e_{13} \ e_{14}]^T$
 $e_2 = [e_{21} \ e_{22} \ e_{23} \ e_{24}]^T$
 $L \rightarrow p \times m$
 $p \rightarrow \text{size of } S$
 $m \rightarrow \text{no. of factors}$
 $S = LL^T + \Psi$
 $\Psi \rightarrow \text{Diagonal matrix } p \times p$

$$\sum_{i=1}^p h_i^2 = \sum_{i=1}^p (l_{i1}^2 + l_{i2}^2 + \dots + l_{im}^2)$$

$$\Psi_i^2 = s_{ii} - h_i^2 = s_{ii} - (l_{i1}^2 + \dots + l_{im}^2)$$

$$L = \begin{bmatrix} \sqrt{\lambda_1} & e_{11} & \sqrt{\lambda_2} & e_{21} \\ & e_{12} & & e_{22} \\ & e_{13} & & e_{23} \\ & e_{14} & & e_{24} \end{bmatrix}_{4 \times 2} \quad L^T = \begin{bmatrix} \sqrt{\lambda_1} e_{11} & e_{12} & e_{13} & e_{14} \\ \sqrt{\lambda_2} e_{21} & e_{22} & e_{23} & e_{24} \end{bmatrix}_{2 \times 4}$$

$$LL^T = \begin{bmatrix} \lambda_1 e_{11}^2 + \lambda_2 e_{21}^2 & \lambda_1 e_{12}^2 + \lambda_2 e_{22}^2 & \lambda_1 e_{13}^2 + \lambda_2 e_{23}^2 & \lambda_1 e_{14}^2 + \lambda_2 e_{24}^2 \\ & & & \\ & & & \\ & & & \end{bmatrix}_{4 \times 4}$$

Now:

$$\sum_{i=1}^p h_i^2 = \sum_{i=1}^4 (\lambda_1 e_{i1}^2 + \lambda_2 e_{i2}^2) = \text{trace}(LL^T)$$

$$= (\lambda_1 e_{11}^2 + \lambda_2 e_{21}^2) + (\lambda_1 e_{12}^2 + \lambda_2 e_{22}^2) + (\lambda_1 e_{13}^2 + \lambda_2 e_{23}^2) + (\lambda_1 e_{14}^2 + \lambda_2 e_{24}^2)$$

$$= \lambda_1 (e_{11}^2 + e_{12}^2 + e_{13}^2 + e_{14}^2) + \lambda_2 (e_{21}^2 + e_{22}^2 + e_{23}^2 + e_{24}^2)$$

We know that components of an eigenvector are normalized

$$\therefore \boxed{e_{i1}^2 + e_{i2}^2 + e_{i3}^2 + e_{i4}^2 = 1}, \quad i=1,2$$

$$\sum_{i=1}^p h_i^2 =$$

$$\lambda_1 + \lambda_2$$

$\rightarrow$ largest $m=2$ eigenvalues

$$= 4 \cdot 2 + 2 \cdot 1$$

$$= \underline{\underline{6 \cdot 3}}$$

Hence proved.

which are λ_3 and λ_2 here in our question.

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

Ans-3: $X_1, \dots, X_n$ random sample of size n .
Mean $\mu \in \mathbb{R}$ and variance μ^2 .

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \quad \left\{ \text{as } \text{Cov}(X_i, X_j) = 0 \text{ for } i \neq j \right\} \\ &= \frac{1}{n^2} \times n \mu^2 = \frac{\mu^2}{n} \end{aligned}$$

We know: $\sqrt{n} (\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$

$$\text{then } \sqrt{n} (g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N(0, [g'(\mu)]^2 \sigma^2)$$

$$\therefore \text{variance of } g(\bar{X}_n) \xrightarrow{t} [g'(\mu)]^2 \times \frac{\mu^2}{n}$$

Now, it should be not depend on μ .

$$\Rightarrow [g'(\mu)]^2 \cdot \mu^2 = c \quad \{\text{constant}\}$$

$$\Rightarrow [g'(\mu)]^2 = \frac{c}{\mu^2} \Rightarrow \boxed{g'(\mu) \propto 1/\mu}$$

Therefore:

$$g'(\mu) = \frac{k}{\mu} \quad (k \text{ is any proportionality constant})$$

Integrating both sides, wrt μ , $g(\mu) = k \ln \mu + c$

05 $\therefore \boxed{g(\mu) = k \ln \mu + c}$ $\leftarrow$ Here k, c are constants.

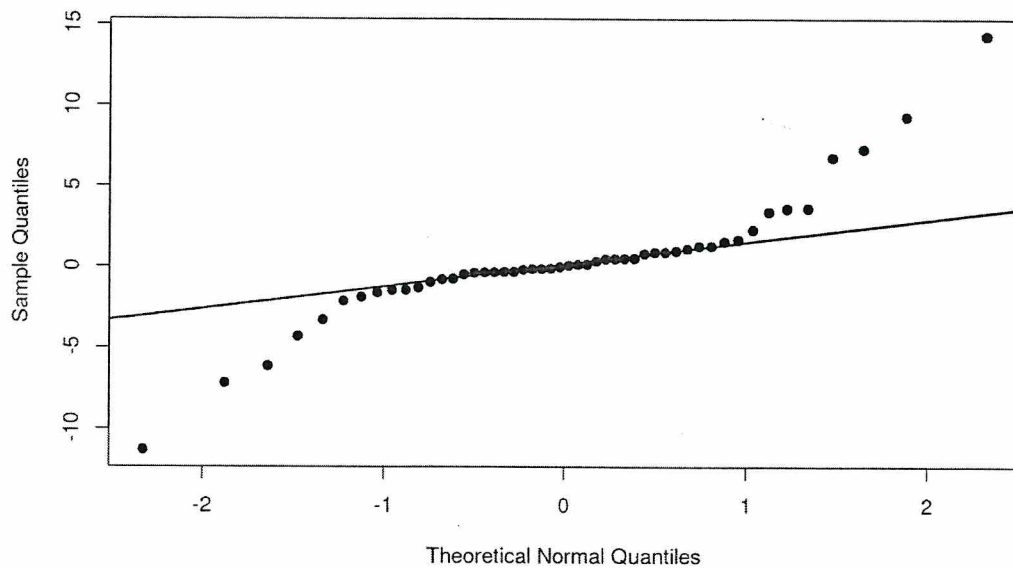
Checking correctness

$$g'(\mu) = \frac{k}{\mu} \Rightarrow \text{Var}(g(\bar{X}_n)) = \frac{[g'(\mu)]^2 \mu^2}{n}$$

$$= \frac{\frac{k^2}{\mu^2} \times \mu^2}{n} = \frac{k^2}{n}$$

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Independent of μ .

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



Ans. 4.

Q-Q Plot for a dataset of $n = 50$

- ① This data does not follow a normal distribution, as it is not the ideal normal curve.
- ② • This type of data & plot will ~~not~~ get characterized to skew form of data.
• It will be lognormal distribution here in plot.)

~~② This:~~

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ADDITIONAL PAGES FOR ANSWERS

